

HOMEWORK 6 FOR MATH 2860 (ELEMENTARY DIFFERENTIAL EQUATIONS)

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Due at 10:00 EST in class

Question Q1. Given $y_1(x) = x^2$ is a solution of the ODE

$$x^2 y'' - 3xy' + 4y = 0,$$

- (a) Find a second solution to the ODE.
- (b) What is the domain/interval on which the two solutions are form a fundamental set of solutions? (In other words, on what domain/interval are the two solutions linearly independent?)
- (c) Find the general solution of the ODE on the interval from part (b).

Show all your work for full credit.

Question Q2. Is $y_p = \frac{1}{2}x + \frac{11}{12}$ a particular solution of the nonhomogeneous equation $y''' - 6y'' - 11y' + 6y = -3x$?

Show all your work for full credit.

Question Q3.

- (a) Show that $2 \cos x = e^{ix} + e^{-ix}$
- (b) Show that $\frac{d}{dx}(\sin x) = \frac{e^{ix} + e^{-ix}}{2}$

Show all your work for full credit.